Curs 1 - Introducere

A system can be any collection of hardware, software, and firmware components.

Metrics = criteria used to evaluate the performance of a system
Workloads = the requests maade by the users of a systems

Why do we need EP?

- Determine a system's capabilities
- Determine feasibility of future expansion
- System comparison
- Identify performance problems/bottlenecks
- System tuning

Process

1. Select appropriate evaluation techniques, performance metrics, system workload
2. Conduct performance measurements
3. Use statistical techniques for comparing alternatives
4. Design simulation/measurement experiments to get maximum results with least effort
5. Perform simulations correctly
6. Mathematical techniques

Main parts

1. Overview of Performance Evaluation
2. Measurement Techniques and Tools
3. Probability Theory and Statistics
4. Experimental Design and Analysis
5. Simulation
6. Queueing Theory

Measurement techniques and tools

- Types of workloads
- Popular benchmarks
- The art of workload selection
- Workload characterization techniques
- Monitors
- Accounting logs
- Monitoring distributed systems
- Load drivers
- Capacity planning
- The art of data presentation
- Ratio games

Probability theory and statistics

- Probability and statistics concepts
- Summarizing measured data by a single number
- Summarizing the variability of measured data
- Graphical methods to determine distributions of measured data
- Sample statistics
- Confidence interval
- Comparing two alternatives
- Measure of relationship
- Simple and multiple linear regression models
- Other regression models

Experimental. design and analysis

- Introduction to experimental design
- 2^k factorial designs
- $2^k r$ factorial designs with replications
- 2^{k-P} fractional factorial designs
- One factor experiments
- Two factors full factorial design without replications
- Two factors full factorial design with replications
- General full factorial designs with *k* factors

Simulation

- Introduction to simulation
- Types of simulation
- Model verification and validation
- Analysis of simulation results
- Random-number generation
- Testing random-number generators
- Random-variate generation
- Commonly used distributions

Example

- What performance metrics should be used to compare the performance of two disk drives?
- Which type of monitor (software or hardware) would be more suitable for measuring each of the following quantities:
  - Number of Instructions executed by a processor?

- Response time of packets on a network?

Example (cont)
The number of packets losts on two links was measured for four files sizes as shown below:

File size	Link A	Link B
50	0	1
1000	5	10
1400	7	3
1800	3	0

Which link is better?

Example (cont)

The performance of a system depends on the following three factors:

- Garbage collection technique used: Gl, G2, or none.
- Type of workload: editing, computing, or Al.
- Type of CPU: Cl, C2, or C3.

How many experiments are needed?

How does one estimate the performance impact of each factor?

Examples (cont)

In order to compare the performance of two cache replacement algorithms:

- What type of simulation model should be used?
- How long should the simulation be run?
- What can be done to get the same accuracy with a shorter run?
- How you decide if the random-number generator in the simulation is a good generator?

The art of performance evaluation

Example:

• These are the throughputs of two systems A and B in transactions per second

System	Workload 1	Workload 2
A	20	10
B	10	20

Possible solutions

1. Compare the average

System	Workload 1	Workload 2	Average
A	20	10	15
B	10	20	15

Conclusion: The two systems are equally good

2. Compare the ratio with system B as the base

System	Workload 1	Workload 2	Average
A	2	0.5	1.25
B	1	1	1

Conclusion: System A is better than B

3. Compare the ratio with system A as the base

System	Workload 1	Workload 2	Average
A	1	1	1
B	0.5	2	1.25

Conclusion: System A is better than B

4. Similar games in: Selection of workload, Measuring the systems, Presenting the results